

Name _____

Date April 15, 2025

me during this test
yikes 1.5

1. Determine $f''(-1)$ for the function $f(x) = 3x^6 + 13x^4 - \frac{9}{x}$.

$$f(x) = 3x^6 + 13x^4 - \frac{9}{x}$$

$$f'(x) = 18x^5 + 52x^3 + \frac{9}{x^2}$$

$$f''(x) = 90x^4 + 156x^2 - \frac{18}{x^3}$$

$$f''(1) = 90(1)^4 + 156(1)^2 - \frac{18}{(1)^3} = 228$$

$$f''(1) = 228$$

2. The concentration of a drug in a patient's bloodstream is $C(t) = -1 + 2t + \frac{8}{t+1}$, where C is in milligrams per cubic centimetre and t is the number of hours after the drug is taken. Find the maximum and minimum concentrations for the interval $0 \leq t \leq 4$.

$$C(t) = -1 + 2t + \frac{8}{t+1}$$

$$C'(t) = 2 + 0 - \frac{8}{(t+1)^2}$$

$$C'(t) = 2 - \frac{8}{(t+1)^2}$$

$$0 = 2 - \frac{8}{(t+1)^2}$$

$$2 = \frac{8}{(t+1)^2}$$

$$\frac{2(t^2 + 2t + 1)}{2} = \frac{8}{2}$$

$$t^2 + 2t - 3 = 0$$

$$t = 1, -3$$

↑ inadmissible

$$C(0) = 7$$

$$C(1) = 5 \rightarrow \text{min}$$

$$C(4) = 8.6 \rightarrow \text{max} \quad (4)$$

the max concentration is 8.6 mg/cm^3 @ $t = 4 \text{ h}$ and the min is 5 mg/cm^3 @ $t = 1$

3. A particle moves along a line. The particle's distance from a fixed point, s , in metres as a function of time, t , in seconds can be modelled by $s(t) = 2t^3 + 3t^2 - 36t + 40$, $t \geq 0$. When is the particle moving in a positive direction?

$$s(t) = 2t^3 + 3t^2 - 36t + 40$$

$$v(t) = 6t^2 + 6t - 36$$

$$= 6(t^2 + t - 6)$$

$$= 6(t-2)(t+3)$$

$$t = 2, -3$$

↑ inadmissible

$$(0, 2) \cup (2, \infty)$$

$v(t) < 0$ $v(t) > 0$

the particle moves in a positive direction when $t > 2$

(2)

4. To the right is a velocity (m/s) graph of a particle travelling along a horizontal line over the interval $[0, 4]$. Are the following statements about the velocity graph true or false? Be sure to justify your response.

a) The particle is moving to the right only during the interval $t < 3$.

FALSE. t is moving ~~to~~ the right $(0, 1)$
AKA when the y -value is $(+)$. t is moving to the left $(1, 3)$ when the y -value is $(-)$.

b) The particle is only slowing down when $0 < t < 1$.

FALSE. For the particle to slow down, the signs of $v(t)$ and $a(t)$ must be opposing. During $(0, 1)$, $v(t) = (+)$ and

$a(t) = (-)$. However, this also happens @ other points in time

5. At what point(s) is the tangent line to the curve $y^3 = 2x^2$ perpendicular to the line $x + 2y - 2 = 0$? like

$$x + 2y - 2 = 0$$

$$1 + 2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{1}{2}$$

$$y^3 = 2x^2$$

$$3y^2 \frac{dy}{dx} = 4x$$

$$\frac{dy}{dx} = \frac{4x}{3y^2}$$

$$\frac{4x}{3y^2} = -\frac{1}{2} \rightarrow \text{PERP.}$$

$$\frac{4x}{3y^2} = -\frac{1}{2}$$

$$4x = -\frac{3}{2}y^2$$

$$x = \frac{by^2}{4}$$

$$x = \frac{3y^2}{2}$$

$$x + 2y - 2 = 0$$

$$2y = -x + 2$$

$$y = -\frac{x}{2} + 1$$

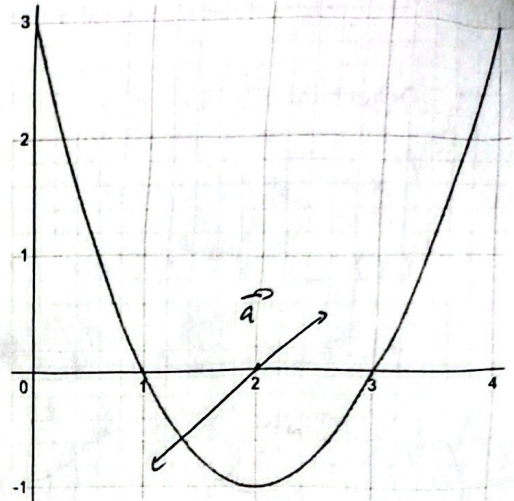
$$y = \left(\frac{3y^2}{2}\right)\left(-\frac{1}{2}\right) + 1$$

$$y = \frac{3y^2}{4} + 1$$

$$\frac{4y}{4} - \frac{3y^2}{4} = 1$$

$$-3y^2 + 4y - 4 = 0$$

$$y = \frac{4 \pm \sqrt{16 - 48}}{-6}$$



$$x = \frac{3y^2}{2}$$

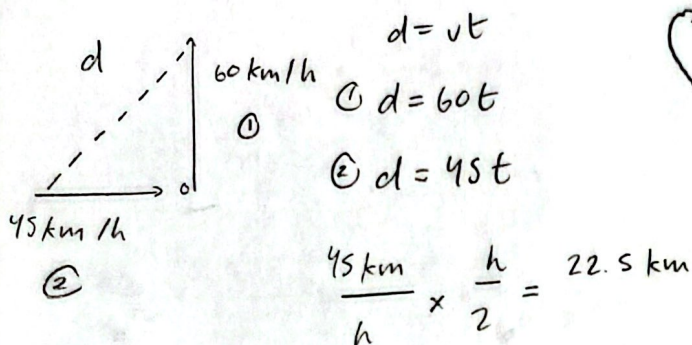
$$x = 3 \left(\frac{4 + \sqrt{13}}{6} \right)^2 \left(\frac{1}{2} \right)$$

$$x = 3 \left(\frac{4 - \sqrt{13}}{6} \right)^2 \left(\frac{1}{2} \right)$$

$$P \left(\frac{29 + 8\sqrt{13}}{24}, \frac{4 + \sqrt{13}}{6} \right)$$

$$P \left(\frac{29 - 8\sqrt{13}}{24}, \frac{4 - \sqrt{13}}{6} \right)$$

6. A bus leaves a bus station at 3pm and heads north at 60 km/h. At the same time, a car that is located 40km west of the bus station, is heading east at 45 km/h. When were the vehicles the closest together if they drive for half an hour? (to the nearest minute). How far apart are they at this time in kilometres? Include a diagram in your solution.



$$d = \sqrt{(22.5 - 45t)^2 + (60t)^2}$$

$$d^2 = (22.5 - 45t)^2 + (60t)^2$$

$$d' = \frac{1}{2} (5625t^2 - 2025t + 506.25)^{-\frac{1}{2}}$$

$$d' = \frac{11250t - 2025}{2 \sqrt{5625t^2 - 2025t + 506.25}}$$

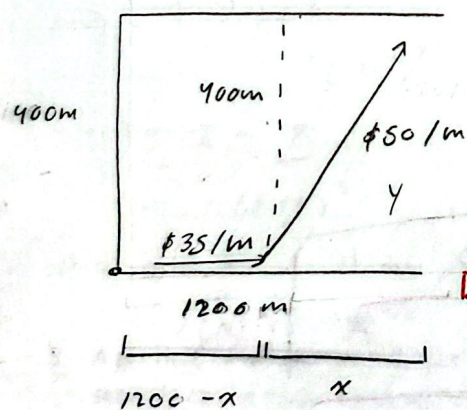
$$0 = \frac{11250t - 2025}{2 \sqrt{5625t^2 - 2025t + 506.25}}$$

$$11250t - 2025 = 0$$

$$t = 0.18 \text{ h} \rightarrow 10.8 \text{ min}$$

$$d(0.18) = 18 \text{ km}$$

7. A power line is needed to connect a power station on the shore of a river to an island 1200 metres downstream and 400 metres offshore. The cost to lay wire underwater is \$50/m and \$35/m to lay it underground. What is the least expensive way to lay the wire and what is the minimum cost?



DOMAIN?

$$C = 42000 - 35x + 400 + 42000 - 35x$$

$$C = 84000 - 70x + 400$$

$$C = 84400 - 70x$$

$$x = 1205.71 \text{ m} \rightarrow y = 8.94$$

$$\text{cost} = 247.15$$

lay 1205.71 m underground and 8.94 m underwater for a \$ of \$247.15

$$(50y)^2 = (400)^2 + (35(1200 - x))^2$$

$$50y = \sqrt{400^2 + (42000 - 35x)^2}$$

$$y = \sqrt{\frac{(400)^2 + (35(1200 - x))^2}{2500}}$$

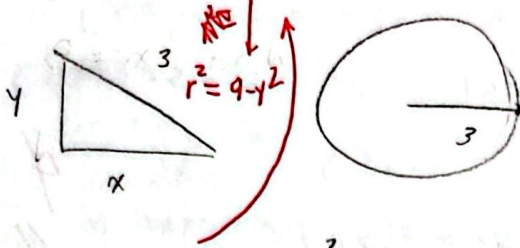
so

$$C = 35(1200 - x) + 50y$$

$$C = 35(1200 - x) + \sqrt{(400)^2 + (35(1200 - x))^2}$$

8. Determine the dimensions of the volume of the largest right circular cone ($V = \frac{1}{3}\pi r^2 h$) that can be inscribed in a sphere of radius 3.

$$V = \frac{1}{3}\pi r^2 h \rightarrow r = x \quad \text{radius} = x$$



$$h = 3 + y \quad 3^2 = y^2 + x^2$$

solve in terms of y

$$y = 3 - h \quad y = \sqrt{(-x)^2 + 3^2}$$

$$h = 3 + y$$

$$h = 3 + \sqrt{3^2 + (-x)^2}$$

$$h = 3 + 3 - x$$

$$V = \frac{1}{3}\pi r^2 h$$

plug $r = x$ and $h = 3 + 3 - x$

$$V = \frac{1}{3}\pi x^2 (6 - x)$$

$$= \frac{2\pi x^2}{3} - \frac{\pi x^3}{3}$$

$$V' = 4\pi x - \frac{3\pi x^2}{3}$$

$$= 4\pi x - \pi x^2$$

$$= x(4\pi - \pi x)$$

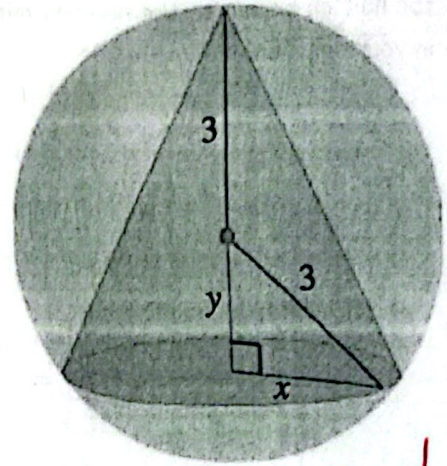
$$\downarrow$$

$$x = 0$$

$$\downarrow$$

$$4\pi - \pi x = 0$$

$$x = 4$$



$$\text{radius} = x$$

$$r = 4$$

$$h = 3 + y$$

$$y = \sqrt{(-4)^2 + 3^2}$$

$$= 5$$

$$h = 3 + y$$

$$h = 8$$

$$\text{height} = 8 \text{ units}$$

$$\text{radius} = 4 \text{ units}$$

Demetri

2.5